

Test cases for vending machine specification.

① Reset mode

$rstn == 0 \rightarrow$ All signals i/p, o/p, ctrl should go to def value as mentioned here in spec as "X".

② configure mode

$rstn == 1, cfg_mode == 1$

Case 1

cfg-mode after rstn occurs:

operation signals like item-valid, currency-value, item-dispense should still have the value "X",

while APB signals are
cfg with address 0 for total available items,

rest address	31	24	23	16	15	0	} for particular address means particular item
	no-of. disp-items		no-of. avail-items		item- price		

Case 2

cfg-mode == 1 between process,

apb same operation in case 1, while operation signals need to hold their values then need to be updated on new values after configured, until "no transaction must not to be done".

also for both case 1 & 2, available count & dispensed count for each item is updated correctly check to be done.

③ operation mode (let empty-item-indication represents item 10)
rstnr == 0 && cfg-mode == 0

→ APB signals pclk, pwrtck, ... all signals must freeze its default cfg value no change should be there,
except available & dispense count in registers.

ex:

item-no	item-name	cost	available	dispensed	empty item indication
5	chocolate	10	5	3	0
8	cake	30	0	10	10

$\Rightarrow$ item-select = item-cfg [8],

item-valid = 1,

currency-value = 10,

currency-valid = 1,

check

empty-item-indication = 10,

currency-change = currency-value.

MAX_ITEMS = 32, so 5 bits for item.

	item-cfg (5)	item-cfg (8)	item-cfg (8)
item-select			
item-valid	1	1	1
currency-value	10	20	50
currency-valid	1	1	1
item-disp valid	1	0	1
item-disp	5'b00101	5'b0	5'b01000
currency-change	5'b0	5'b01010	5'b10100

④ available-items ≤ 0 ,

currency-valid, item-valid = 1,

currency-value = ~~10~~ 10,

check

empty-item-indication == 10 ?

↳ item no use left

⑤ currency-value = {5, 10, 15, 20, 25, 100}



currency-valid = 0.

⑥ available-items != 0 $\xrightarrow{T}$ item-dispense-valid = 1,
 $\downarrow F$ item-dispense = item-cfg[-],

item-dispense-valid = 0,

item-dispense = 0,

currency-change = currency-value

currency-change



> than

change

= currency-value
 - item price

change
 = 0.

⑦ item-dispense-valid high for only one clk. (sys clk)

⑧ After currency-valid = 1'b1, within 10 sys clk, o/p should be dispensed.

⑨ $prstn = 0$, all ~~clk~~ signals except clock should go to def value:

⑩ After each item dispensed, 'pdata's available and dispensed count need to be handled.